

Engineering Mechanics

Lecture 6

By

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Structure

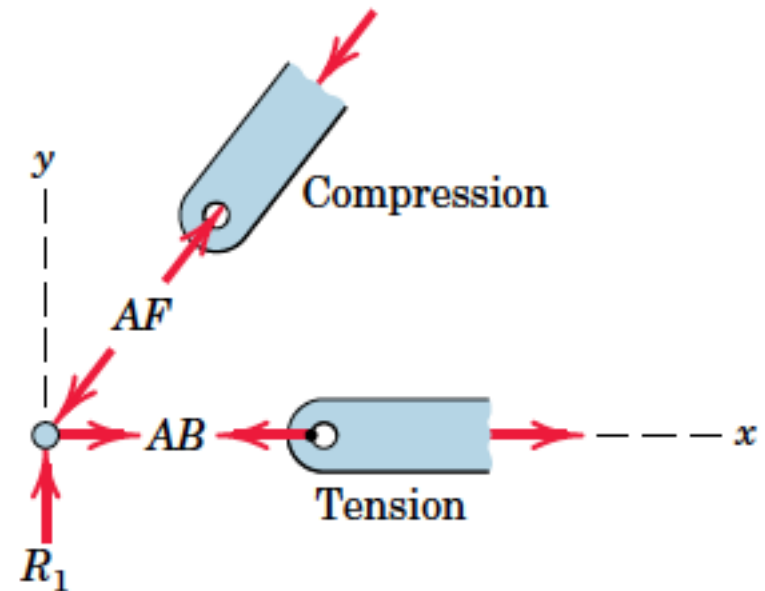
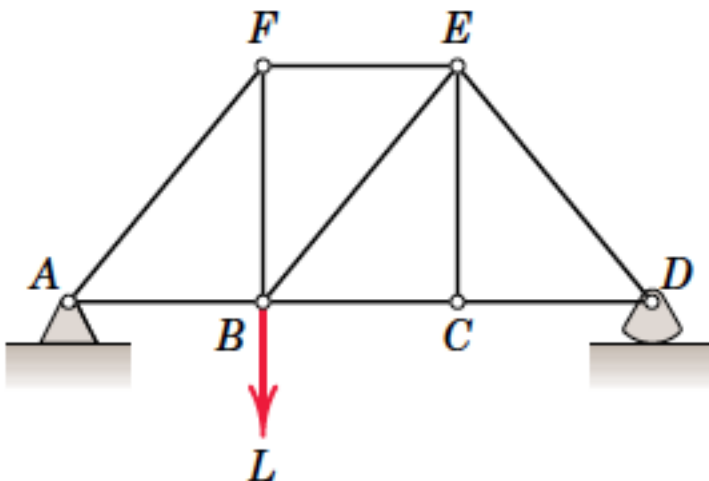
- Any **connected system of members** built to **support or transfer forces** and to safely **withstand the loads** applied to it.
- Type of Structures
 - Truss
 - Frame
 - Machine
- In this lecture, we consider only **statically determinate** structures

Trusses



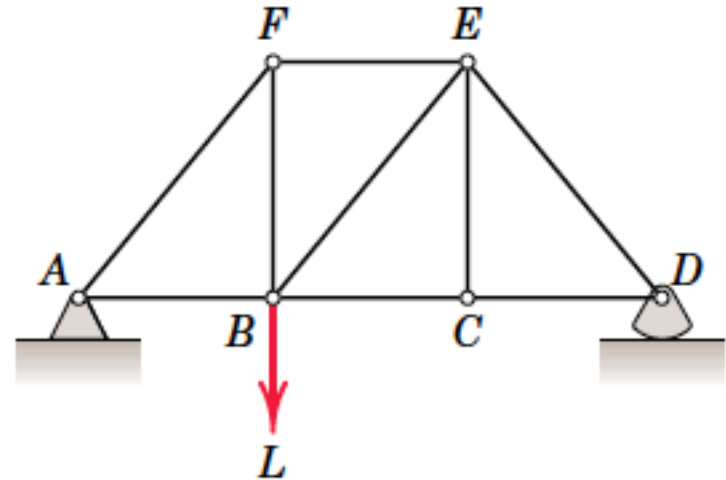
Internal forces

- In this discussion we focus on the determination of the forces **internal to a structure**
- that is, forces of action and reaction between the **connected members**.

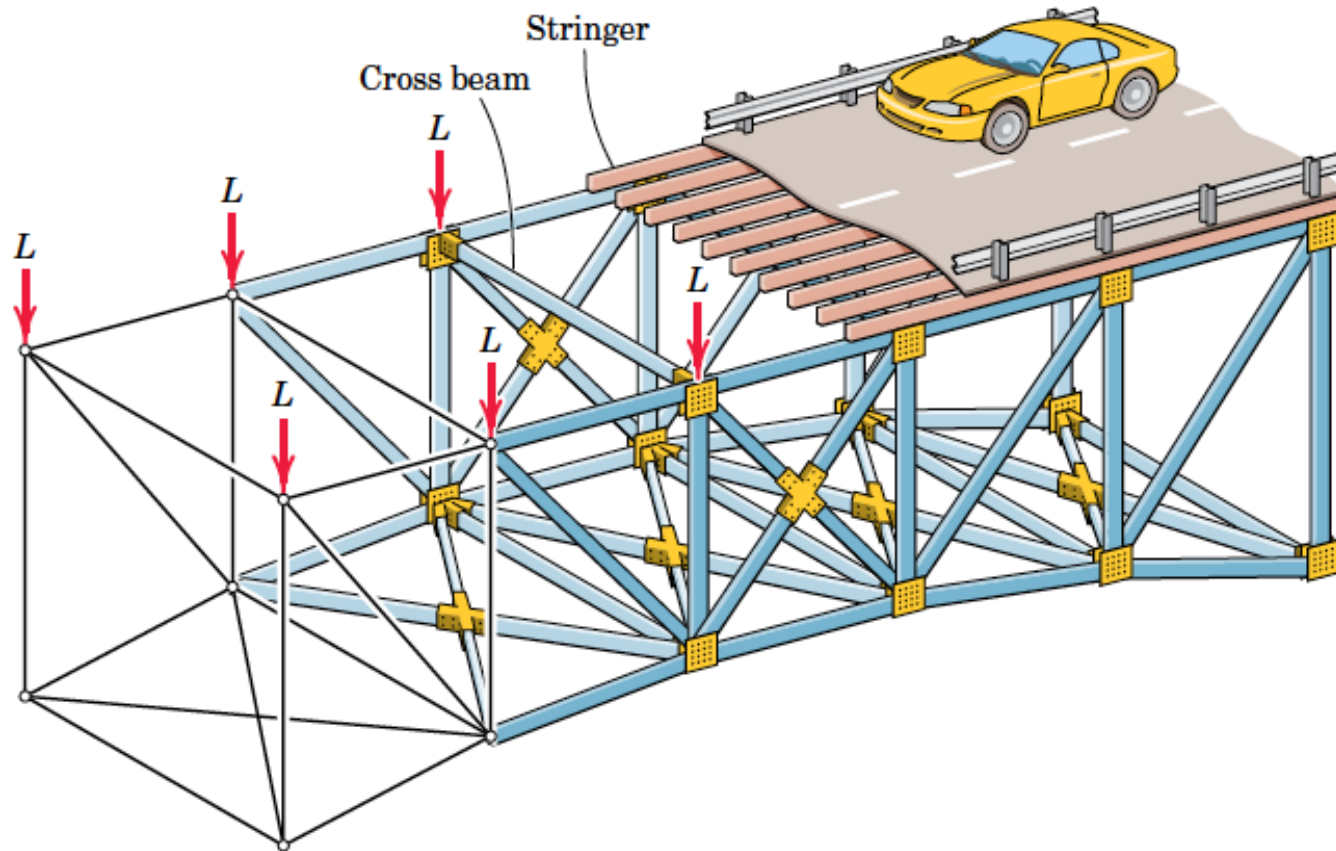


Plane Truss

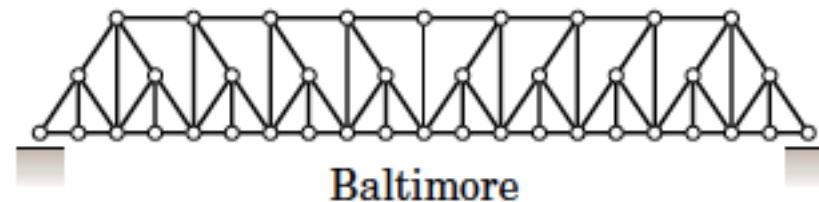
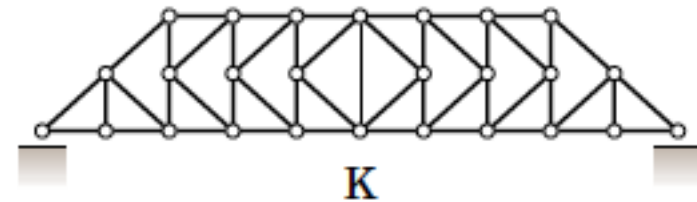
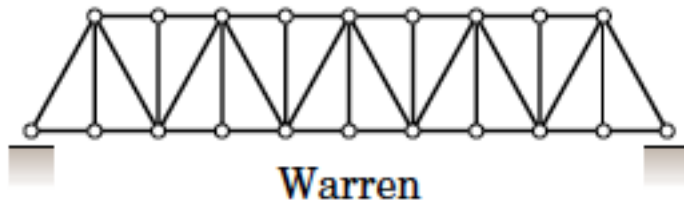
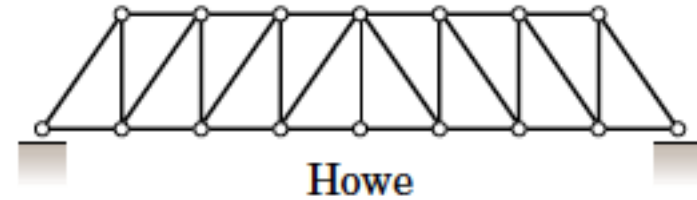
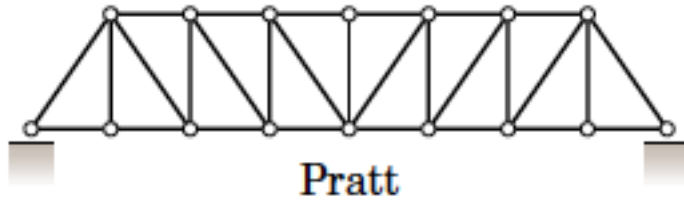
- A framework composed of **members joined at their ends** to form a rigid structure is called a truss.
- Examples: **Bridges, roof supports, derricks**, and other such structures



Section of a typical bridge structure

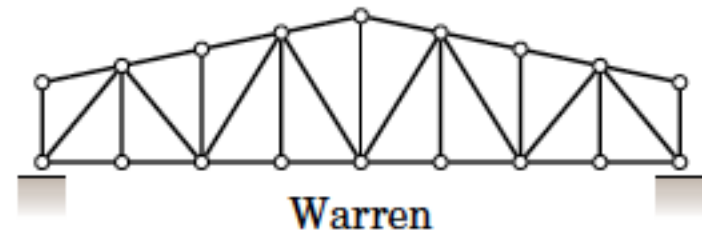
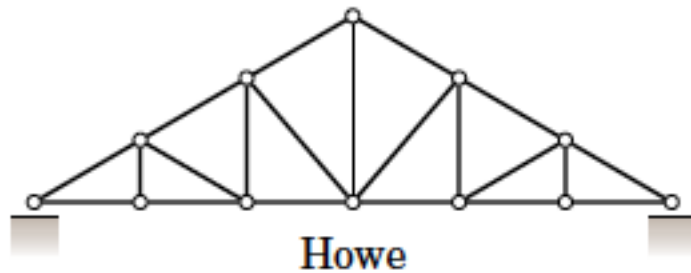
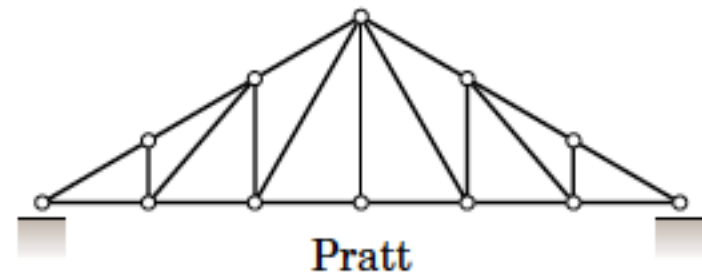
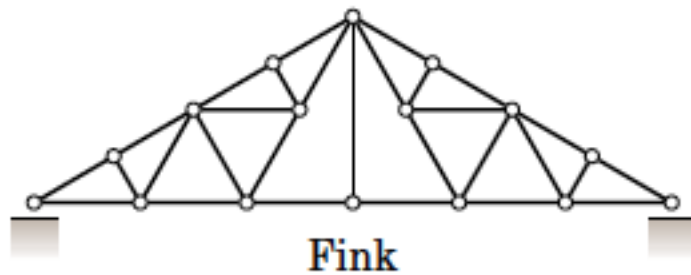


Commonly Used Bridge Trusses



Commonly Used Bridge Trusses

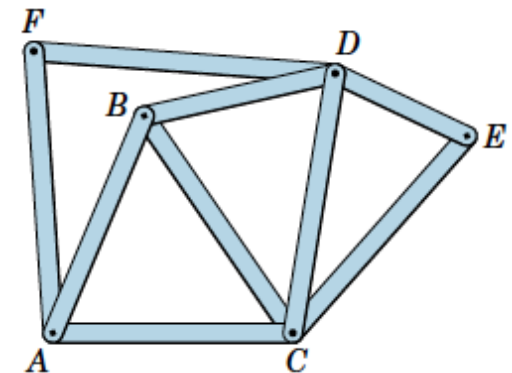
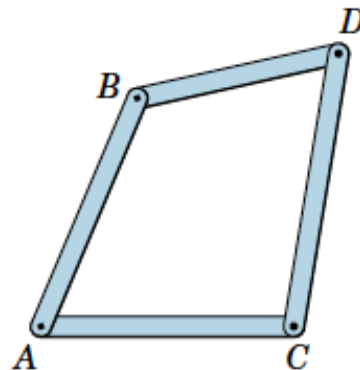
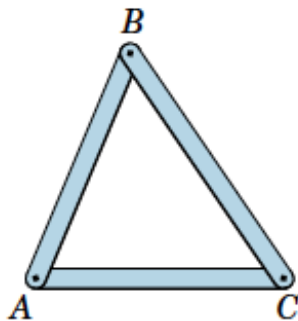
Commonly Used Roof Trusses



Commonly Used Roof Trusses

Simple Trusses

- The basic element of a plane truss is **the triangle**. It constitute a **rigid frame**.
- **Four or more bars** pin-jointed to form a polygon of as many sides constitute a **nonrigid frame**
- We can make nonrigid frame stable, **by adding a diagonal bar** joining A and D or B and C and thereby forming two triangles.
- Structures built from a **basic triangle** in the manner described are known as **simple trusses**.

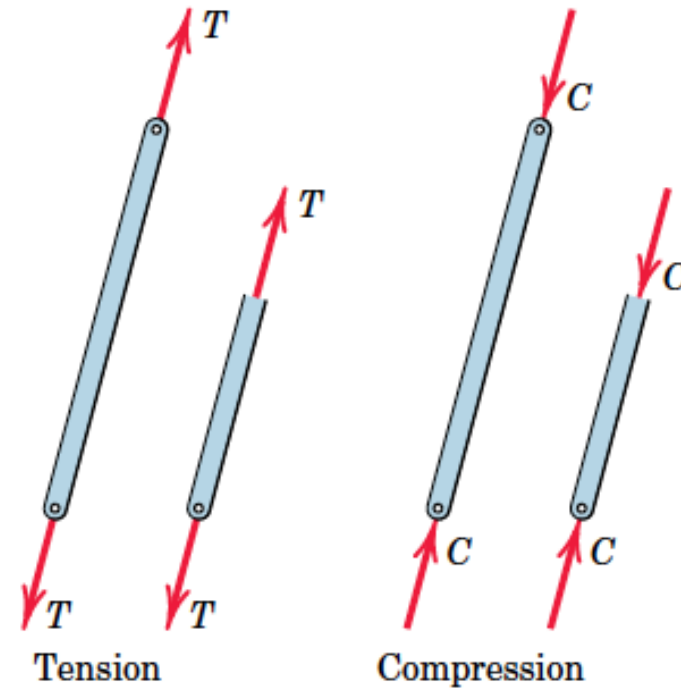


Design of a Truss

- First **determine the forces** in the various members and then
- Select **appropriate sizes and structural shapes** to withstand the forces.
- This lecture focuses on **determine the forces in the various members**

Assumption

- All members to be two-force members
- The two forces are equal, opposite, and collinear for equilibrium
- The member may be in **tension** or **compression**



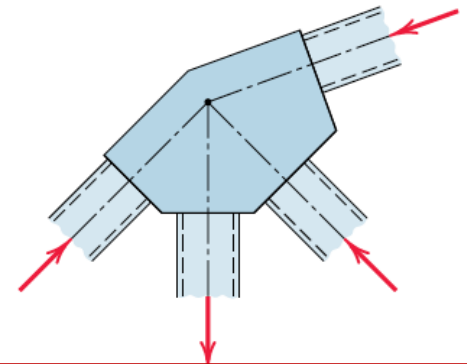
Two-Force Members

Assumption

- **Weight** of the member is **small** compared with the force it supports.
- If not we can **replace the weight W** of the member by **two forces, each $W/2$** if the member is uniform, each end of the member
- This gives the **correct result for the average tension or compression** along the member but will **not account for the effect of bending** of the member.

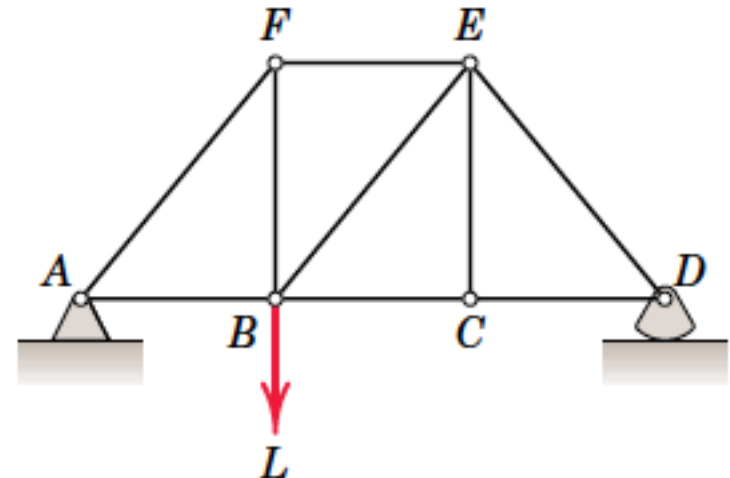
Truss Connections and Supports

- All **external forces** are applied at the pin connections
- When **welded or riveted connections** are used to join structural members,
- we may usually assume that the connection is a **pin joint** if the centrelines of the members are **concurrent** at the joint



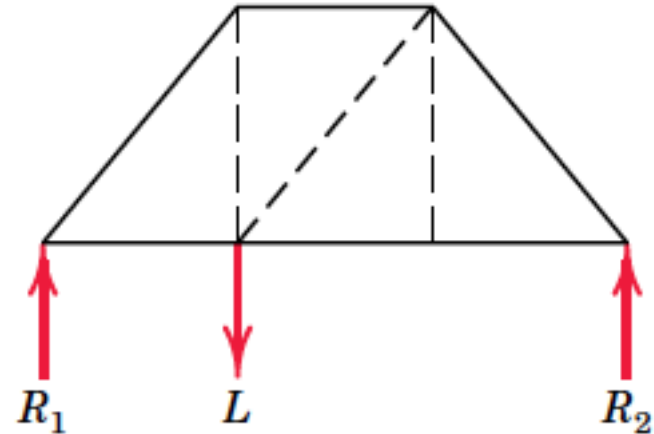
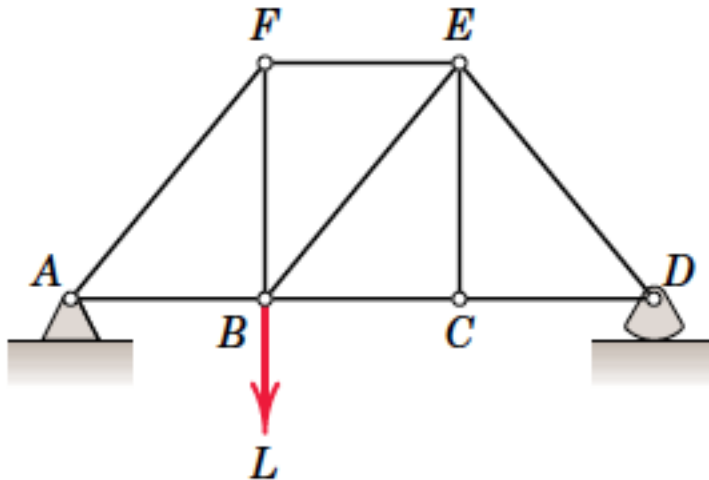
Truss Connections and Supports

- A **roller**, or some kind of **slip joint** is used at one of the supports to provide **for expansion and contraction due to temperature changes and for deformation from applied loads**.
- Trusses and frames in which no such provision is made are **statically indeterminate**



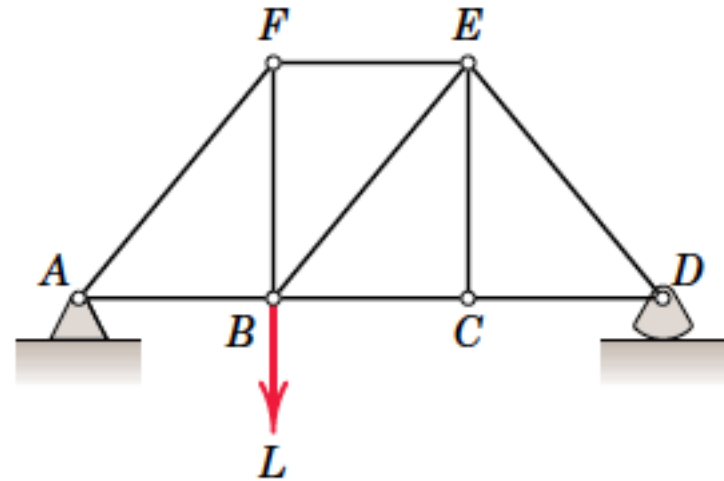
Force Analysis of Simple Truss

- Step 1 : Find external reactions by applying the equilibrium equations to the truss as a whole.



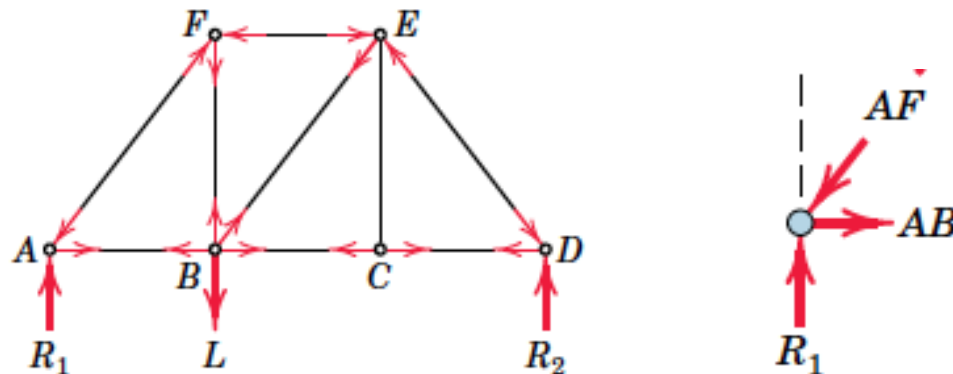
Force Analysis of Simple Truss

- Step 2 : Finding the forces in the members of a truss
 - Method of joint or
 - Method of section



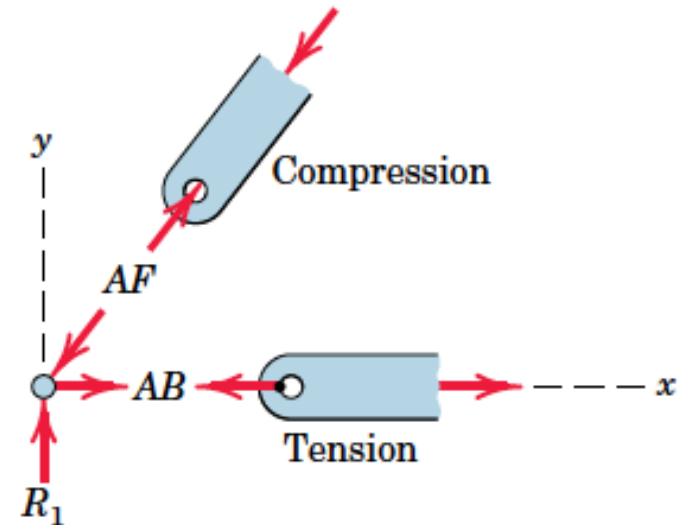
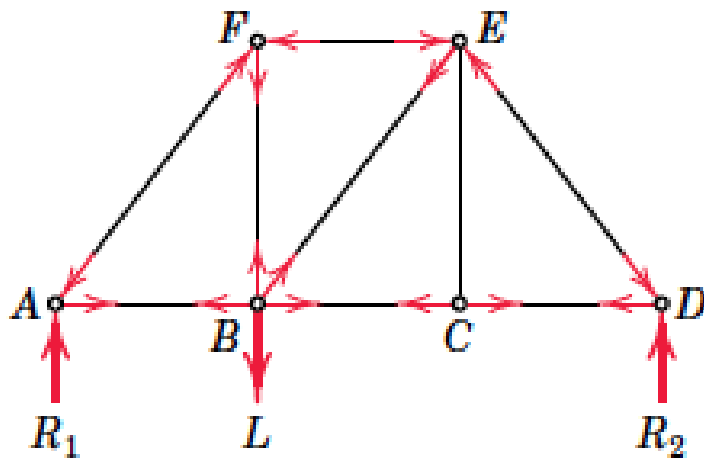
Method of Joint

- It consists of satisfying the conditions of equilibrium for the forces acting on the **connecting pin of each joint**.
- The method therefore deals with the **equilibrium of concurrent forces**, and
- only **two independent** equilibrium equations are involved.

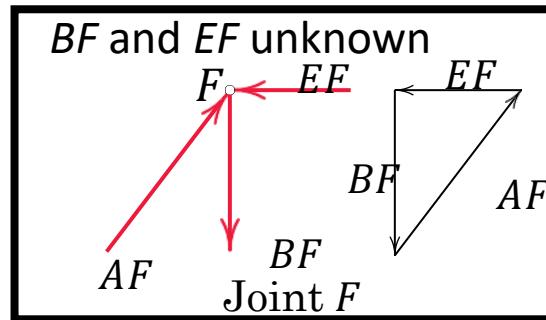
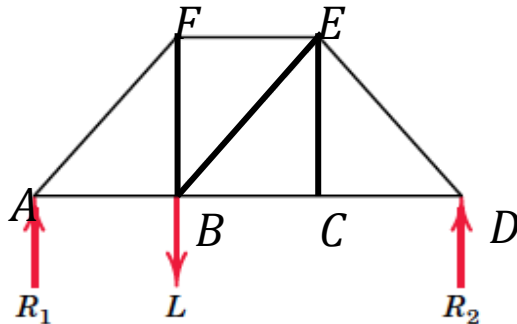


Method of Joints

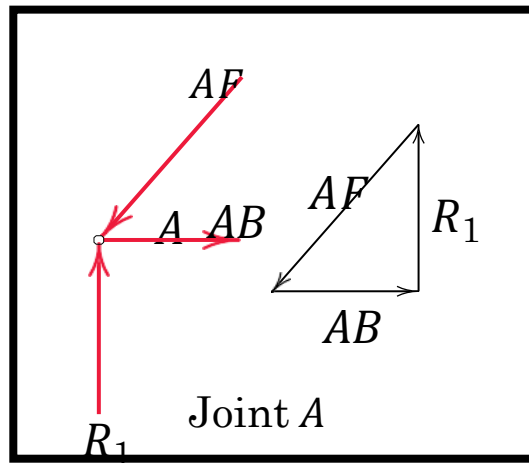
- We begin the analysis with any joint where at least **one known load** exists and where **not more than two unknown forces** are present.
- The solution may be started with the **pin at the left end**.



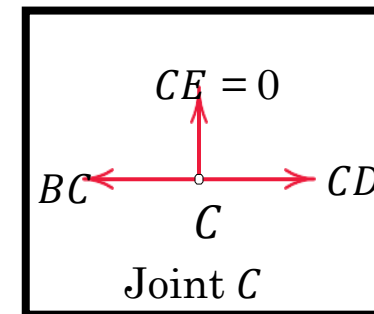
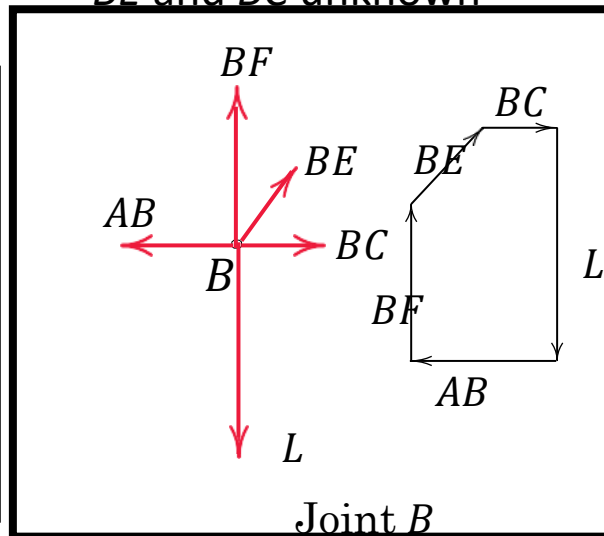
Approach: Solving for Joints A, F, B, C



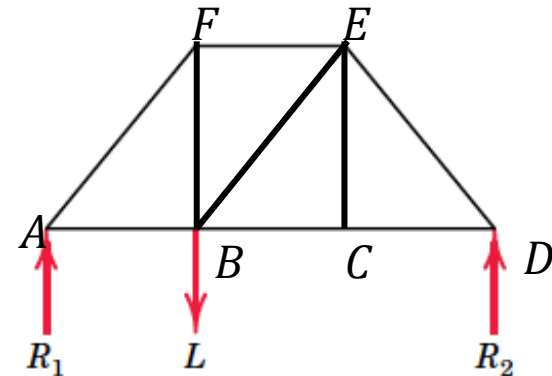
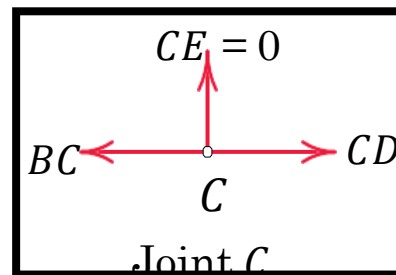
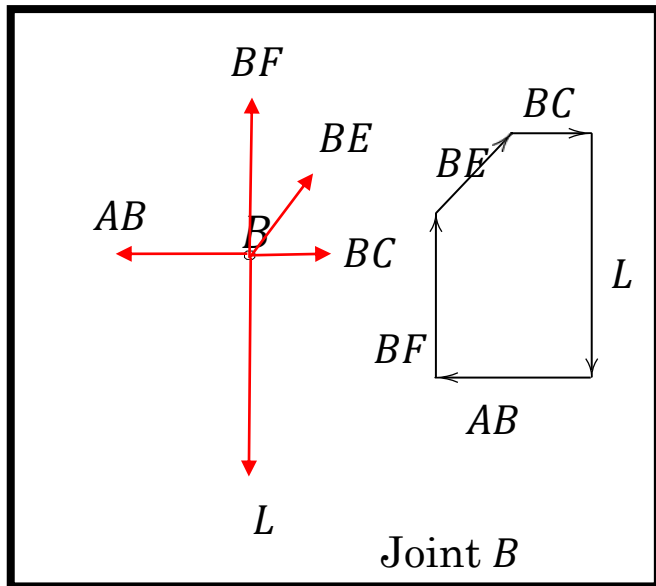
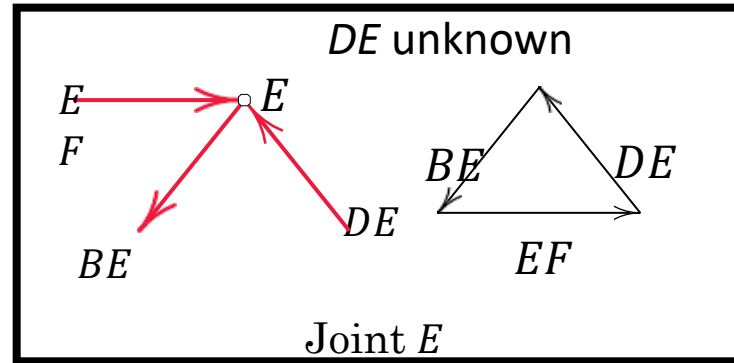
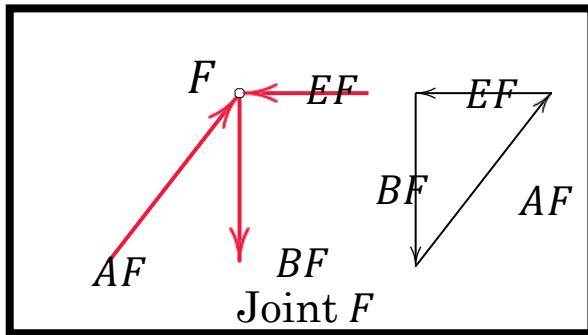
AF and AB unknown



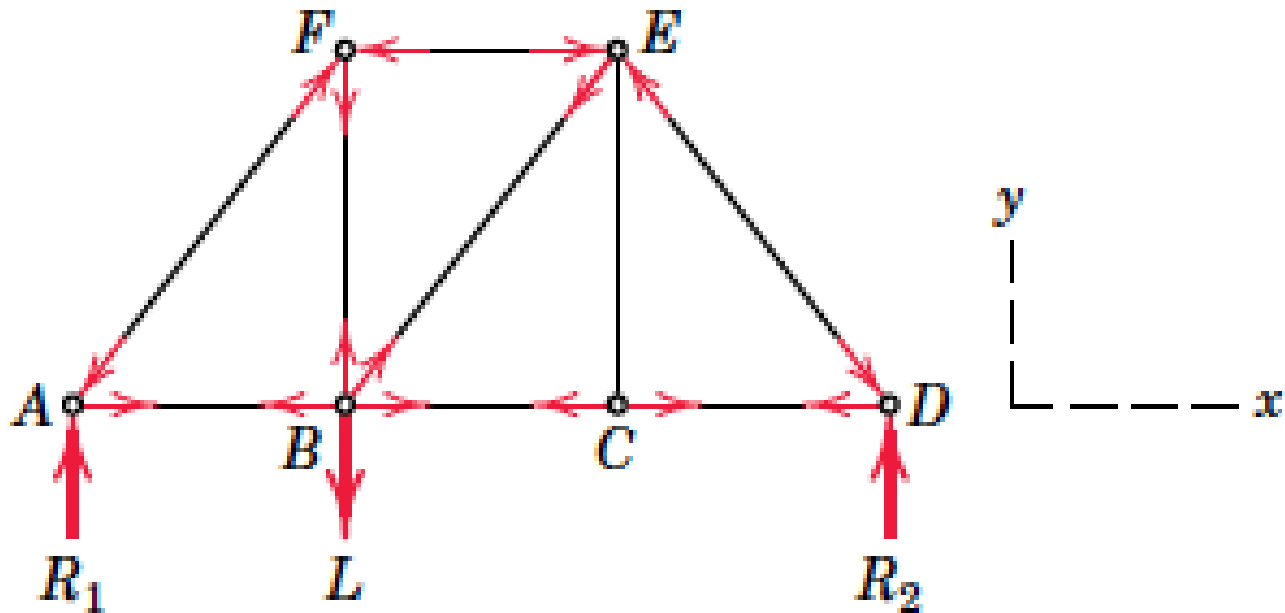
BE and BC unknown



Approach : Solving for Joint E



Approach: Forces at Joints



Internal and External Redundancy

Truss is Statically indeterminate when

1. **More external supports** than are necessary to ensure equilibrium,
 - the extra supports constitute external redundancy.
2. **More internal members** than are necessary to prevent collapse then the
 - extra members constitute internal redundancy and the truss is again

Condition for determinacy

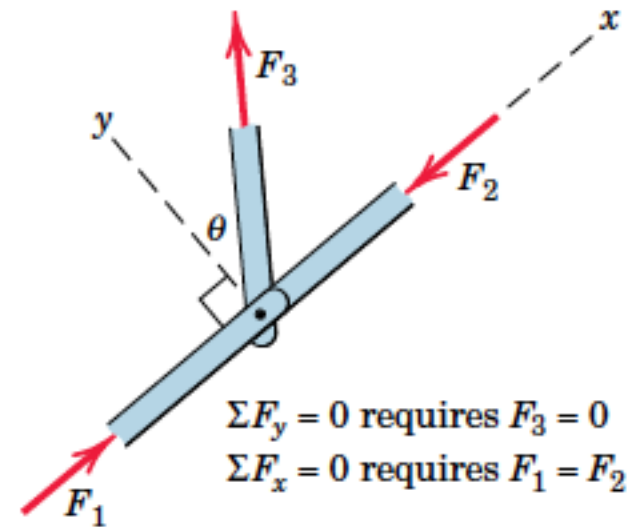
- Equilibrium of each joint by 2 equations hence for j joints total $2j$ equations
- truss composed of m two-force members and having the maximum of three unknown support reactions, there are in all $m + 3$ unknowns
- For any plane truss, the equation $m + 3 = 2j$ will be satisfied if the truss is statically determinate internally.

Internal Redundancy

- For triangle $m=3$, $j=3$ and $m + 3 = 2j = 6$
- If $m + 3 > 2j$, there are more members than independent equations, and the truss is statically indeterminate internally
- If $m + 3 < 2j$, there is a deficiency of internal members, and the truss is unstable and will collapse under load.

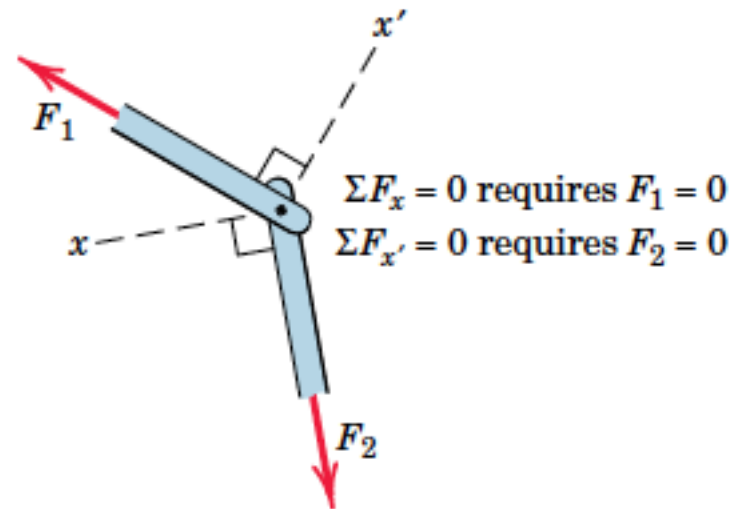
Special Cases

- When **two collinear** members are **under compression**,
- it is necessary to **add a third member** to maintain alignment of the two members and prevent buckling.
- F_3 in the third member is zero
- $F_1 = F_2$.



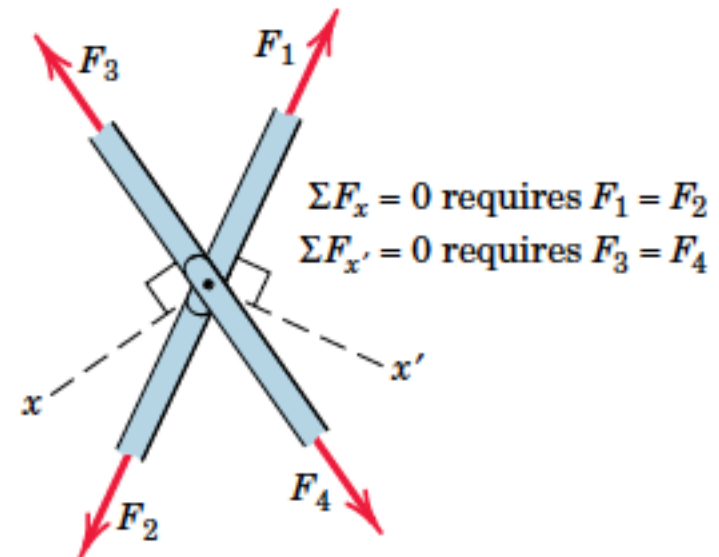
Special Cases

- When **two noncollinear members are joined** and no externally applied load at this joint,
- the **forces in both members must be zero**,



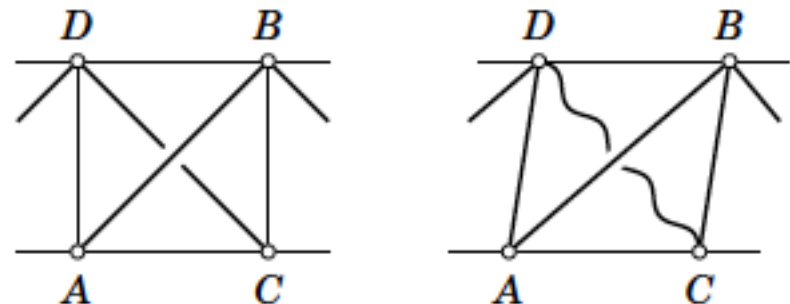
Special Case

- two pairs of collinear members are joined
- the forces in each pair must be equal and opposite



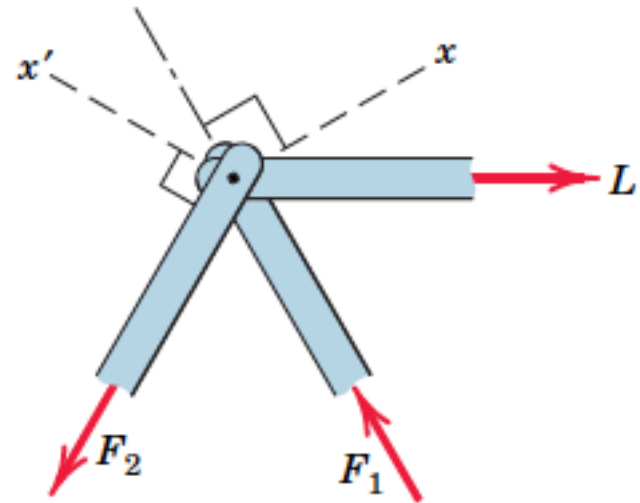
Special cases

- Truss panels are frequently cross-braced
- Such a panel is statically indeterminate if each brace can support either tension or compression.
- For flexible members capable of supporting compression, then only the tension member acts and we can disregard the other member.



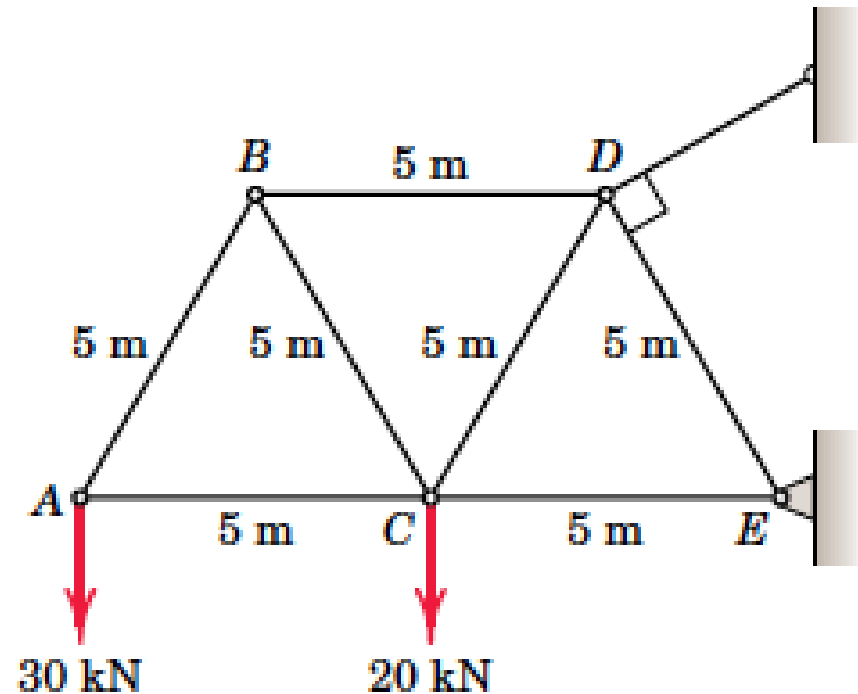
Choice of frame

- We can avoid **simultaneous solution** of the equilibrium equations for two unknown forces at a joint by a **careful choice of reference axes**

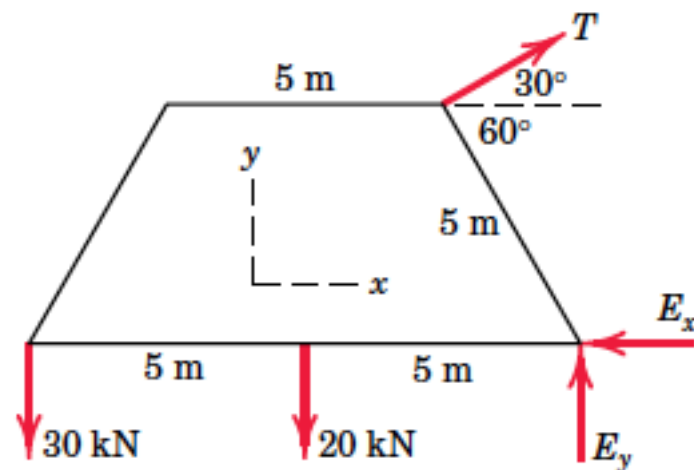
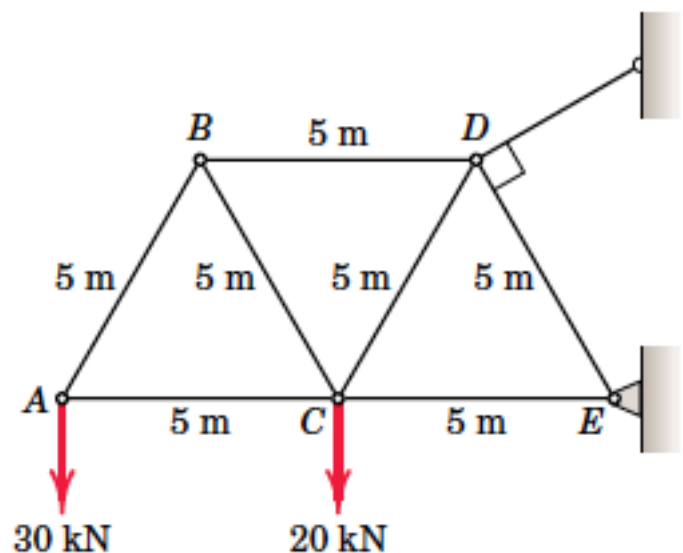


Example 1

- Compute the force in each member of the loaded cantilever truss by the method of joints.



Example 1: External reactions



$$[\Sigma M_E = 0]$$

$$5T - 20(5) - 30(10) = 0$$

$$T = 80 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$80 \cos 30^\circ - E_x = 0$$

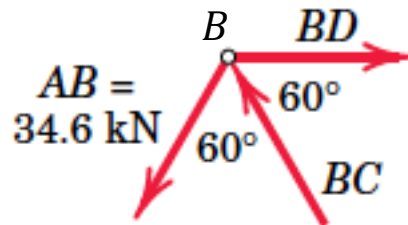
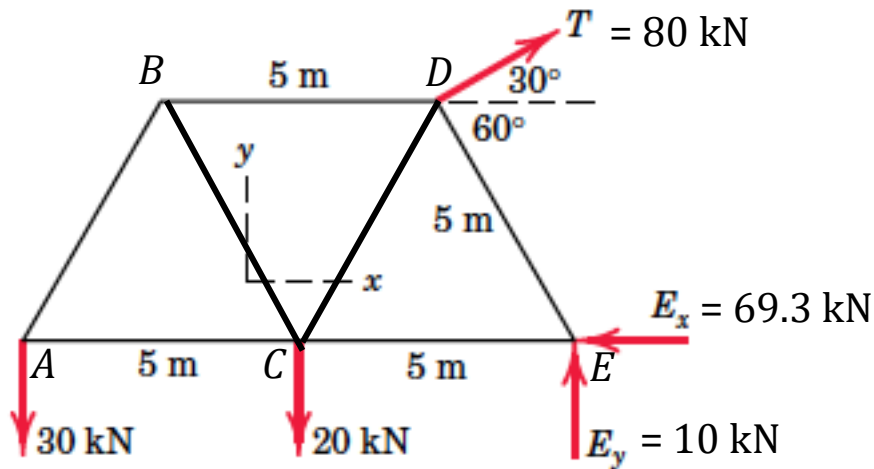
$$E_x = 69.3 \text{ kN}$$

$$[\Sigma F_y = 0]$$

$$80 \sin 30^\circ + E_y - 20 - 30 = 0$$

$$E_y = 10 \text{ kN}$$

Example 1: Forces in Members



$$[\Sigma F_y = 0]$$

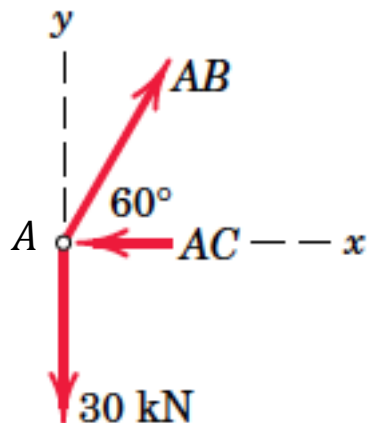
$$0.866AB - 30 = 0$$

$$AB = 34.6 \text{ kN T}$$

$$[\Sigma F_x = 0]$$

$$AC - 0.5(34.6) = 0$$

$$AC = 17.32 \text{ kN C}$$



$$[\Sigma F_y = 0]$$

$$0.866BC - 0.866(34.6) = 0$$

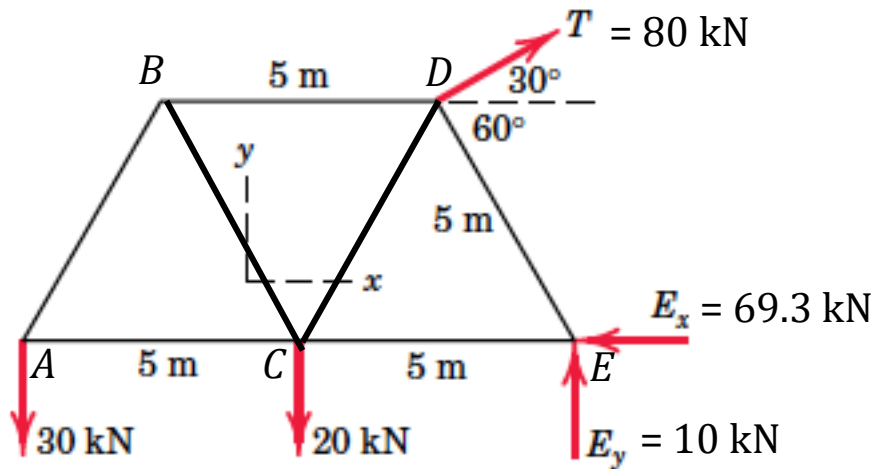
$$BC = 34.6 \text{ kN C}$$

$$[\Sigma F_x = 0]$$

$$BD - 2(0.5)(34.6) = 0$$

$$BD = 34.6 \text{ kN T}$$

Example 1: Forces in Members



$$[\Sigma F_y = 0]$$

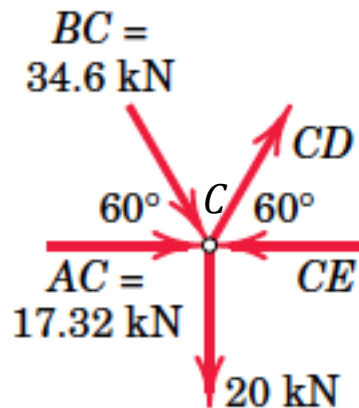
$$0.866CD - 0.866(34.6) - 20 = 0$$

$$CD = 57.7 \text{ kN T}$$

$$[\Sigma F_x = 0]$$

$$CE - 17.32 - 0.5(34.6) - 0.5(57.7) = 0$$

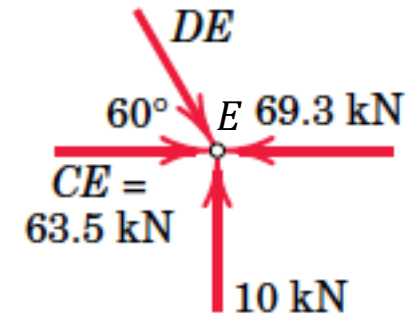
$$CE = 63.5 \text{ kN C}$$



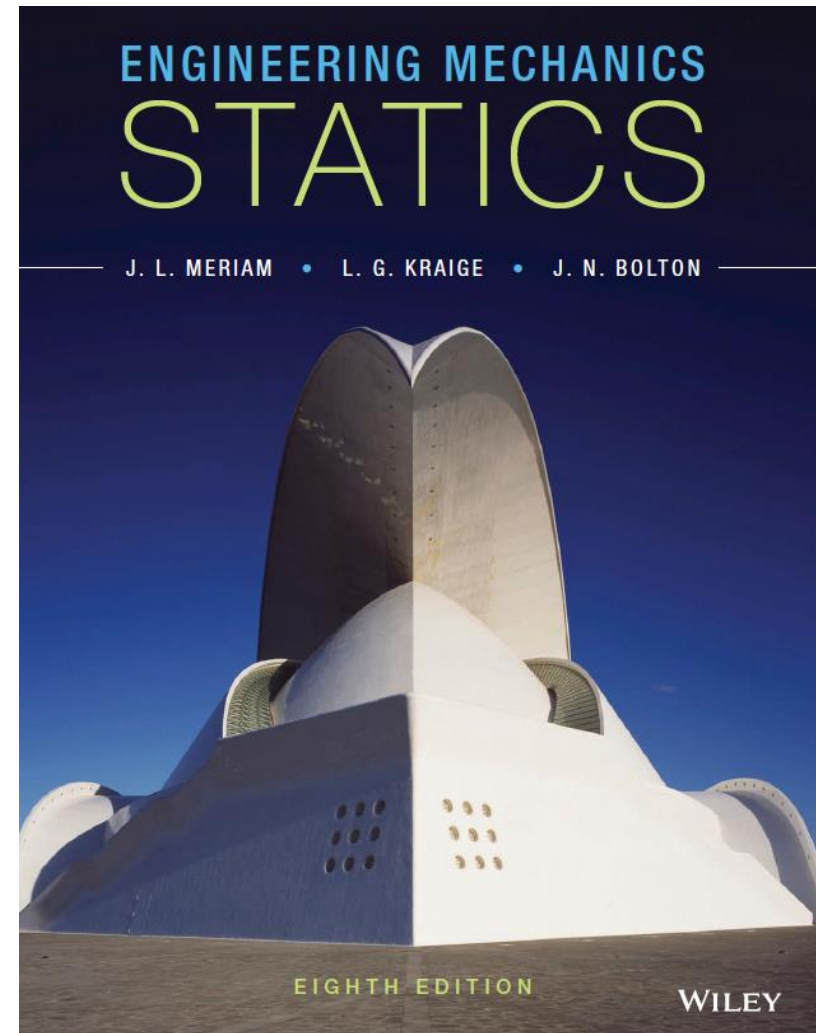
$$[\Sigma F_y = 0]$$

$$0.866DE = 10$$

$$DE = 11.55 \text{ kN C}$$



References :



THANK YOU